

Department Mapping (Department.java)

```
package com.cognizant.ormlearn.model;

import java.util.Set;

import javax.persistence.Column;
import javax.persistence.Entity;
import javax.persistence.FetchType;
import javax.persistence.Id;
import javax.persistence.OneToMany;
import javax.persistence.Table;

@Entity
@Table(name = "department")
public class Department {

    @Id
    @Column(name = "dp_id")
    private int id;

    @Column(name = "dp_name")
    private String name;

    @OneToMany(mappedBy = "department", fetch = FetchType.LAZY)
    private Set<Employee> employeeList;
```

```

public Department() {}

public int getId() { return id; }

public void setId(int id) { this.id = id; }

public String getName() { return name; }

public void setName(String name) { this.name = name; }

public Set<Employee> getEmployeeList() { return employeeList; }

public void setEmployeeList(Set<Employee> employeeList)
{ this.employeeList = employeeList; }
}

```

Skill Mapping(Skill.java)

```

package com.cognizant.ormlearn.model;

import javax.persistence.Column;
import javax.persistence.Entity;
import javax.persistence.Id;
import javax.persistence.Table;

@Entity
@Table(name = "skill")

public class Skill {

```

```
@Id

@Column(name = "sk_id")

private int id;


@Column(name = "sk_name")

private String name;


public Skill() {}


public int getId() { return id; }

public void setId(int id) { this.id = id; }


public String getName() { return name; }

public void setName(String name) { this.name = name; }


@Override

public String toString() {

    return "Skill [id=" + id + ", name=" + name + "]";

}

}
```

Employee Mapping (Employee.java)

```
package com.cognizant.ormlearn.model;
```

```
import java.math.BigDecimal;
```

```
import java.util.Date;
```

```
import java.util.Set;
```

```
import javax.persistence.Column;
```

```
import javax.persistence.Entity;
```

```
import javax.persistence.FetchType;
```

```
import javax.persistence.Id;
```

```
import javax.persistence.JoinColumn;
```

```
import javax.persistence.JoinTable;
```

```
import javax.persistence.ManyToMany;
```

```
import javax.persistence.ManyToOne;
```

```
import javax.persistence.Table;
```

```
import javax.persistence.Temporal;
```

```
import javax.persistence.TemporalType;
```

```
@Entity
```

```
@Table(name = "employee")
```

```
public class Employee {
```

```
    @Id
```

```
    @Column(name = "em_id")
```

```
    private int id;
```

```
@Column(name = "em_name")
```

```
private String name;
```

```
@Column(name = "em_salary")
```

```
private BigDecimal salary;
```

```
@Column(name = "em_permanent")
```

```
private boolean permanent;
```

```
@Column(name = "em_date_of_birth")
```

```
@Temporal(TemporalType.DATE)
```

```
private Date dateOfBirth;
```

```
@ManyToOne(fetch = FetchType.LAZY)
```

```
@JoinColumn(name = "em_dp_id")
```

```
private Department department;
```

```
@ManyToMany(fetch = FetchType.LAZY)
```

```
@JoinTable({
```

```
    name = "employee_skill",
```

```
    joinColumns = @JoinColumn(name = "es_em_id"),
```

```
    inverseJoinColumns = @JoinColumn(name = "es_sk_id")
```

```
})
```

```
private Set<Skill> skillList;
```

```
public Employee() {}
```

```
public int getId() { return id; }
```

```
public void setId(int id) { this.id = id; }
```

```
public String getName() { return name; }
```

```
public void setName(String name) { this.name = name; }
```

```
public BigDecimal getSalary() { return salary; }
```

```
public void setSalary(BigDecimal salary) { this.salary = salary; }
```

```
public boolean isPermanent() { return permanent; }
```

```
public void setPermanent(boolean permanent) { this.permanent = permanent; }
```

```
public Date getDateOfBirth() { return dateOfBirth; }
```

```
public void setDateOfBirth(Date dateOfBirth) { this.dateOfBirth = dateOfBirth; }
```

```
public Department getDepartment() { return department; }
```

```
public void setDepartment(Department department) { this.department =  
department; }
```

```
public Set<Skill> getSkillList() { return skillList; }
```

```
public void setSkillList(Set<Skill> skillList) { this.skillList = skillList; }
```

```

@Override

public String toString() {

    return "Employee [id=" + id + ", name=" + name + ", salary=" + salary + ",
permanent=" + permanent + "];"

}

}

```

Repository with Optimized JOIN FETCH HQL (EmployeeRepository.java)

```

package com.cognizant.ormlearn.repository;

import java.util.List;

import org.springframework.data.jpa.repository.JpaRepository;
import org.springframework.data.jpa.repository.Query;
import org.springframework.stereotype.Repository;
import com.cognizant.ormlearn.model.Employee;

@Repository

public interface EmployeeRepository extends JpaRepository<Employee, Integer>
{

    @Query("SELECT e FROM Employee e LEFT JOIN FETCH e.department d LEFT
JOIN FETCH e.skillList WHERE e.permanent = true")

    List<Employee> getAllPermanentEmployees();

}

```

Service Implementation (EmployeeService.java)

```

package com.cognizant.ormlearn.service;

```

```
import java.util.List;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;
import com.cognizant.ormlearn.model.Employee;
import com.cognizant.ormlearn.repository.EmployeeRepository;
```

```
@Service
```

```
public class EmployeeService {
```

```
    @Autowired
```

```
    private EmployeeRepository employeeRepository;
```

```
    @Transactional(readOnly = true)
```

```
    public List<Employee> getAllPermanentEmployees() {
```

```
        return employeeRepository.getAllPermanentEmployees();
```

```
    }
```

```
}
```

Execution Runner File (OrmLearnApplication.java)

```
package com.cognizant.ormlearn;
```

```
import java.util.List;
```

```
import org.slf4j.Logger;
```



```
import org.slf4j.LoggerFactory;

import org.springframework.boot.SpringApplication;

import org.springframework.boot.autoconfigure.SpringBootApplication;

import org.springframework.context.ApplicationContext;

import com.cognizant.ormlearn.model.Employee;

import com.cognizant.ormlearn.service.EmployeeService;
```

```
@SpringBootApplication
```

```
public class OrmLearnApplication {
```

```
    private static final Logger LOGGER =
    LoggerFactory.getLogger(OrmLearnApplication.class);
```

```
    private static EmployeeService employeeService;
```

```
    public static void main(String[] args) {
```

```
        ApplicationContext context =
        SpringApplication.run(OrmLearnApplication.class, args);
```

```
        LOGGER.info("Inside main");
```

```
        employeeService = context.getBean(EmployeeService.class);
```

```
        testGetAllPermanentEmployees();
```

```
    }
```

```
    public static void testGetAllPermanentEmployees() {
```

```
        LOGGER.info("Start");
```

```
List<Employee> employees =
employeeService.getAllPermanentEmployees();

LOGGER.debug("Permanent Employees:{}", employees);

employees.forEach(e -> LOGGER.debug("Skills:{}", e.getSkillList()));

LOGGER.info("End");

}

}
```